



Vossie Trading Hall

The student-only campus marketplace for Eduvos. Buy, sell and trade safely with verified **@vossie.net** students.

PRESENTED BY

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MODULE

ITECA3-12

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THE PROBLEM

Students have no safe place to trade

Eduvos students constantly buy and sell textbooks, electronics and services, but the tools they use today were never built for a closed campus community.



Public marketplaces are open to outsiders

Facebook Marketplace and Gumtree have no student verification, which opens the door to scams, no-shows and unsafe meet-ups.



WhatsApp groups are disorganised

Listings vanish in the chat scroll. There are no categories, no search and no easy way to find what you need.



No proof a deal happened

Nothing records that an item changed hands, which leaves disputes and uncertainty around the hand-over.



THE SOLUTION

A verified, student-only marketplace

Vossie Trading Hall lets verified Eduvos students list items, browse by category, and confirm hand-overs with a unique QR code, so every trade stays inside the campus community.

**Student-only access**

Registration is gated to @vossie.net emails.

**Organised and searchable**

Categories, search and a dedicated textbook hub.

**QR-confirmed hand-off**

Each completed deal leaves a record.

**Simple and free**

Runs on free-tier Azure hosting.



FUNCTIONAL REQUIREMENTS

What the app does

**Student registration**

Sign up and log in with @vossie.net only.

**Browse and search**

Filter listings across nine categories.

**Textbook marketplace**

Dedicated hub with ISBN, author and edition.

**Scan to confirm**

Buyer scans the QR to confirm a transaction.

**Admin role**

Moderate content and manage promotions.

**Manage listings**

Create, edit and delete listings with photo upload.

**Listing types**

Fixed-price sales and wanted requests.

**Generate listing QR**

Each listing gets a unique seller QR code.

**My Listings dashboard**

Manage everything you have posted in one place.

**Business listings**

Flag for campus sellers and services.



NON-FUNCTIONAL REQUIREMENTS

Quality attributes

**Security**

Bcrypt hashing (cost 12), CSRF tokens, domain-gated sign-up, session regeneration and HTTPS-only.

**Performance**

Lightweight PHP with indexed MySQL queries that stay comfortable on free-tier hosting.

**Usability**

Responsive Tailwind UI with mobile camera QR scanning and a manual fallback.

**Portability**

Docker Compose locally, MySQL and SQLite schemas, and one-click Azure deploy.

**Integrity**

Foreign-key constraints, unique email and QR tokens, and immutable transaction records keep data consistent.



TECHNOLOGY STACK

Built layer by layer

**BACKEND**

PHP 8.2 running the server-side logic over PDO

**DATABASE**

MySQL 8.0, with a SQLite schema for lightweight runs

**FRONTEND**

HTML, Tailwind CSS, JavaScript, including the html5-qrcode scanner

**HOSTING**

Azure App Service on Linux (Free F1 tier) with HTTPS enforced

**CONTAINERS**

Docker and Docker Compose for a reproducible local environment

**VERSION CONTROL**

Git and GitHub for source history and collaboration



SYSTEM ARCHITECTURE

A modular PHP application



Shared includes

config.php (site config, allowed domain, categories), auth.php (sessions, CSRF, login guards), db.php (PDO connection), and the header and footer layout.



Page scripts

index, register, login, create-listing, my-listings, listing, books and admin, each a focused entry point.

QR TRANSACTION FLOW



listing-qr.php shows the seller QR



scan-transaction.php reads it



confirm-transaction.php records the deal



DATABASE SCHEMA

Four related tables

**users**

Accounts: name, email, password_hash, phone and is_admin.

**listings**

Items: title, price, category, image, qr_token and status. Belongs to a user.

**transactions**

Confirmed deals that link a listing to its seller and buyer with a timestamp.

**promotions**

Admin-managed campus promos shown on the home page.

Relationships: one user to many listings, one listing to many transactions, and each transaction references a seller and a buyer. ON DELETE CASCADE keeps records consistent.



KEY FEATURES DEMO

Screens to walk through



Home / Browse



Create a Listing



Listing + Seller QR



Scan and Confirm



Textbook Hub



Admin Dashboard



SECURITY HIGHLIGHTS

Safety built in

**Students only**

Only @vossie.net emails can register.

**Hashed passwords**

Bcrypt (cost 12), never stored in plain text.

**CSRF protection**

Tokens guard every form submission.

**Session safety**

Session ID regenerated on login.

**HTTPS and secure cookies**

Enforced HTTPS with HttpOnly, SameSite cookies.

**Safe uploads**

Images only, max 5MB, validated by MIME type.



LIVE DEMO

See it running



LIVE URL

<https://vossiehallphp75163.azurewebsites.net/>



GITHUB

github.com/xd-Park3rr/EduvosMarket



ACCESS

Register with any @vossie.net address. Admin features via admin@vossie.net.



A complete, secure web application

Vossie Trading Hall demonstrates authentication, full CRUD, image uploads, role-based admin and a QR-driven transaction flow on a portable, containerised, cloud-deployed stack.

Real-world considerations: student verification builds trust, database constraints protect data integrity, and free-tier hosting keeps the platform sustainable for a campus community.